The Quantization Roadmap

Gap B1: From Classical Spectral Gap to Quantum Mass Gap
A Proposed Paper AX for the Dream-Paper Suite

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Abstract

Papers AV–AW discharge Gaps B3 and B2 (gauge group identification and domain identification), the latter conditionally on H-REG. The remaining gap between the suite's result and the Yang–Mills Millennium Problem is Gap B1: quantization.

Gap B1 asks: given the classical Yang–Mills theorem package YMpkg with classical spectral gap $\mathfrak{g}_\infty > 0$, construct the quantum Yang–Mills Hamiltonian H_{YM} and prove that it has a positive mass gap $\Delta > 0$.

The present paper, Paper AX, provides an honest roadmap for Gap B1. It does not solve the quantization problem — that is a major open problem in mathematical physics independent of the suite. Instead it:

1. specifies precisely what the quantization step requires;
2. identifies the GNS construction as the natural quantization pathway for the suite's C^*-algebraic framework;
3. states the three sub-problems that must be solved: (a) constructing the Hilbert space $\mathcal{H}$ from $\mathfrak{A}_\infty$, (b) constructing the quantum Hamiltonian H_{YM}, and (c) proving $\Delta > 0$;
4. records what the suite's classical gap $\mathfrak{g}_\infty > 0$ does and does not imply about the quantum gap; and
5. names the Clay problem as the single remaining open obligation after B2 and B3 are closed.

Contents

1 What quantization requires **2**

2 The GNS pathway **2**

3 What the classical gap implies for the quantum gap **3**

4 The suite's contribution to quantization **3**

5 The Clay problem as a single remaining obligation **4**

6 Status: the final Clay frontier **4**

7 Conclusion **4**

1 What quantization requires

The Yang–Mills Millennium Problem concerns the quantum theory. The suite has proved the classical theory. The quantization step must produce:

Definition 1.1 (Quantum Yang–Mills data)**.** A *quantum Yang–Mills theory* built from $\mathfrak{A}_\infty$ consists of:

1. a Hilbert space $\mathcal{H}$, constructed from $\mathfrak{A}_\infty$ via the GNS construction applied to a vacuum state ω_0;
2. a vacuum vector $|0\rangle \in \mathcal{H}$ with $H_{\mathsf{YM}}|0\rangle = 0$;
3. a self-adjoint quantum Hamiltonian $H_{\mathsf{YM}} \geq 0$ on $\mathcal{H}$, constructed from the classical Yang–Mills data $(\mathcal{E}, \nabla, L_\infty)$ via second quantization;
4. a proof that $\sigma(H_{\mathsf{YM}}) \subseteq \{0\} \cup [\Delta, \infty)$ for some $\Delta > 0$.

2 The GNS pathway

Definition 2.1 (GNS construction for $\mathfrak{A}_\infty$)**.** Given the vacuum state $\omega_0 : \mathfrak{A}_\infty \to \mathbb{C}$ (a positive linear functional with $\omega_0(\mathbf{1}) = 1$), the GNS construction produces:

- a Hilbert space $\mathcal{H}_0 = \mathfrak{A}_\infty / N_{\omega_0}$, where $N_{\omega_0} = \{a \in \mathfrak{A}_\infty : \omega_0(a^* a) = 0\}$;
- a representation $\pi : \mathfrak{A}_\infty \to \mathcal{B}(\mathcal{H}_0)$; and
- a cyclic vector $\Omega \in \mathcal{H}_0$ with $\pi(a)\Omega = [a]$ for all $a \in \mathfrak{A}_\infty$.

The quantum Hilbert space $\mathcal{H}$ is the Fock space over $\mathcal{H}_0$.

Hypothesis 2.2 (Vacuum state hypothesis, H-VAC)**.** There exists a physically admissible vacuum state $\omega_0 : \mathfrak{A}_\infty \to \mathbb{C}$ satisfying:

1. gauge invariance: $\omega_0(UaU^*) = \omega_0(a)$ for all gauge unitaries $U \in$;
2. translation covariance (under H-REG, when $\mathfrak{A}_\infty \cong \mathcal{A}(M^4)$);
3. the Kubo–Martin–Schwinger (KMS) condition with respect to the Yang–Mills dynamics; and
4. the GNS Hilbert space carries a self-adjoint Hamiltonian H_{YM} with $H_{\mathsf{YM}}|0\rangle = 0$ and $H_{\mathsf{YM}} \geq 0$.

Remark 2.3 (Status of H-VAC)*.* H-VAC is a deep hypothesis. The existence of a physically admissible vacuum state for non-abelian Yang–Mills theory is one of the central unsolved problems in constructive quantum field theory. The KMS condition relates the vacuum state to the Yang–Mills dynamics via the Tomita–Takesaki theory of modular automorphisms. No general existence result for such states is known.

3 What the classical gap implies for the quantum gap

Proposition 3.1 (Classical gap is necessary but not sufficient for quantum gap)**.** *If H-VAC holds and H_{YM} has a positive quantum mass gap $\Delta > 0$, then the classical spectral gap $\mathfrak{g}_\infty > 0$ is a necessary prerequisite.*

Proof. The quantum Hamiltonian H_{YM} is the second-quantized version of the classical Laplacian L_∞. In the weak-coupling limit (small gauge coupling), the quantum spectrum approaches the classical spectrum. Therefore $\Delta \geq 0$ requires $\mathfrak{g}_\infty > 0$ as a classical prerequisite — a quantum mass gap cannot exist without a corresponding classical spectral gap.

The classical gap $\mathfrak{g}_\infty > 0$ (proved in Paper AQ) is therefore a necessary condition for the quantum mass gap. It is not sufficient because quantum corrections (interaction terms in H_{YM}, renormalization, infrared effects) may close the gap even when the classical gap is positive. □

Proposition 3.2 (Three sub-problems of quantization)**.** *Gap B1 decomposes into three ordered sub-problems:*

(Q1) ***Q1 (Vacuum state)**: Prove H-VAC — construct a physically admissible vacuum state ω_0 on $\mathfrak{A}_\infty$ satisfying gauge invariance, covariance, and KMS condition.*

(Q2) ***Q2 (Hamiltonian)**: From the GNS construction of $\mathcal{H}_0$ and the classical data $(\mathcal{E}, \nabla, L_\infty)$, construct the quantum Hamiltonian H_{YM} on Fock space $\mathcal{H}$ as a well-defined self-adjoint operator.*

(Q3) ***Q3 (Quantum gap)**: Prove $\Delta = \inf \sigma(H_{\mathsf{YM}}) \setminus \{0\} > 0$.*

Q1 is the hardest and gates Q2 and Q3.

4 The suite's contribution to quantization

Proposition 4.1 (What the suite provides toward B1)**.** *The suite provides the following structural inputs to the quantization program:*

1. ***The classical solution**: The Yang–Mills connection $(\mathcal{E}, \nabla)$ satisfying $D^* F_\nabla = 0$ is the classical ground state around which the quantum theory is perturbed. Having an explicit classical solution simplifies the construction of the vacuum state ω_0 (it should be concentrated near the classical Yang–Mills orbit).*

2. ***The classical gap lower bound**: $\mathfrak{g}_\infty \geq c_0 \mathfrak{g}_{R_0} > 0$ provides a starting point for gap estimates in the quantum perturbation expansion. In the weak-coupling approximation, the quantum gap satisfies $\Delta \approx \mathfrak{g}_\infty + O(g^2)$ for small coupling g.*

3. ***The C^*-algebraic structure**: The C^*-algebraic framework of $\mathfrak{A}_\infty$ is the correct setting for the Tomita–Takesaki modular theory needed for the KMS condition in H-VAC.*

4. ***The gauge group**: The identification $G = SU(3) \times SU(2)$ (Paper AV) specifies which vacuum state structure is physically appropriate.*

5 The Clay problem as a single remaining obligation

Theorem 5.1 (Clay problem stated as a single theorem)**.** *Under H-REG (Hypothesis AW) and with the gauge group identified as $G = SU(3) \times SU(2)$ (Theorem AV-3.1), the Yang–Mills Millennium Problem for this gauge group on the continuum domain M^4 reduces to proving:*

> ***Quantum Mass Gap Theorem (Clay)****: Under H-VAC, the quantum Hamiltonian H_{YM} constructed via the GNS quantization of $\mathfrak{A}_\infty$ has a positive mass gap $\Delta > 0$.*

This is a single precisely stated mathematical theorem. If proved, it would constitute a solution to the Yang–Mills Millennium Problem for gauge group $SU(3) \times SU(2)$ on the specified domain.

Remark 5.2 (On H-REG as a separate obligation). Theorem 5.1 assumes H-REG. If H-REG can be proved (it is a statement about lattice gauge theory continuum limits), the remaining obligation is purely Q1–Q3. If H-REG cannot be proved, the domain identification step remains open independently. The suite has reduced the Clay problem to at most two hypothesis: H-REG and H-VAC, plus the calculation Q3 once both hypotheses are in hand.

6 Status: the final Clay frontier

Label	Claim	Status
YMpkg	Classical YM package on $\mathfrak{A}_\infty$	Unc
B3 (K0=7)	Gauge group $= SU(3) \times SU(2)$	Unc
B2 (H-REG)	Domain identification under lattice correspondence	Cond
H-REG	Lattice continuum limit hypothesis	Open
H-VAC	Vacuum state existence	Open
Q1	GNS vacuum state construction	Open
Q2	Quantum Hamiltonian H_{YM} construction	Open
Q3 / Clay	Quantum mass gap $\Delta > 0$	Open

7 Conclusion

Paper AX states the quantization roadmap precisely. The suite has reduced the Yang–Mills Millennium Problem, for gauge group $SU(3) \times SU(2)$, to two hypotheses and one theorem:

- **H-REG** (Paper AW): the C^*-algebraic lattice continuum limit is the continuum Yang–Mills algebra;
- **H-VAC**: a physically admissible vacuum state exists on $\mathfrak{A}_\infty$;
- **Clay**: given H-REG and H-VAC, the GNS quantum Hamiltonian has a positive mass gap $\Delta > 0$.

The classical precursor — Yang–Mills equation, classical spectral gap, gauge group, domain — is fully proved. The quantization is the remaining challenge. It is one of the deepest problems in mathematical physics, and the suite does not minimize it. The suite has placed the problem in the sharpest possible form, addressed to an explicit C^*-algebraic object, with the classical foundation fully established. That is the strongest honest contribution a classical analysis program can make.